Multimorbidity-Aware In-ICU Sepsis Prediction Using Patient Deterioration Trajectories

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Code References

WHAT ARE WE SOLVING?

- **Sepsis**, a leading cause of death, costs Americans **~$5.6B /yr.**
- Clinicians are **unable to trust** current predictive models.
- **Unintuitive** clinical decision support system software.

DATA

- **MIMICIV** - A de-identified **ICU EHR Database** with 90,000+ patient profiles.
- Patient vitals, procedures, SOFA scores, and **multimorbidity groups** were used for sepsis predictions.

DECISION SUPPORT SYSTEM

- Clinician-facing **in-ICU Sepsis Decision Support Tool**
- **Timely alerts** for evolving patient conditions
- Similar patient profiles and outcomes to **de-bias clinicians**
- **Multimorbidity-based** predictions for rapid clinical decisions

MANAGEMENT VIEW

PATIENTS (43)	AGE	GENDER	RACE	CARE UNIT	TIME SINCE ADMISSION (HH:MM)	SEPSIS PROBABILITY
Jason Xu	63	Male	Portuguese	Trauma SICU (TSICU)	01:25	21.87%
Hikaru Isayama	20	Male	Hispanic/Latino - G...	Trauma SICU (TSICU)	05:49	26.77%
Ethan Vo	59	Male	White	Coronary Care Unit (CCU)	02:25	35.00%
Yash Patel	62	Female	White	Medical Intensive Care Unit (MICU)	08:18	31.29%
Gloria Ye	58	Female	Black/African Ameri...	Medical/Surgical Intensive Care Unit (MICU/SICU)	02:49	18.39%
Utkarsh Lohia	61	Male	White	Medical Intensive Care Unit (MICU)	16:34	57.43%
Susan Walker	82	Female	White	Medical Intensive Care Unit (MICU)	11:49	22.26%
Optimus Prime	64	Female	White	Coronary Care Unit (CCU)	03:27	20.86%
Kate Zhou	85	Female	White	Medical Intensive Care Unit (MICU)	12:15	59.31%
Richard Lewis	61	Female	White	Cardiac Vascular Intensive Care Unit (CVICU)	08:12	60.80%
Samuel Mahjouri	91	Male	White	Medical Intensive Care Unit (MICU)	19:55	34.49%
Joseph Hall	21	Female	White	Medical/Surgical Intensive Care Unit (MICU/SICU)	18:33	19.33%
Jessica Allen	59	Male	White	Medical Intensive Care Unit (MICU)	18:56	34.75%
Thomas Young	60	Male	White	Coronary Care Unit (CCU)	06:37	13.36%
Sarah King	61	Male	Multiple Race/Ethni...	Medical/Surgical Intensive Care Unit (MICU/SICU)	17:36	51.24%
Karen Scott	83	Female	Other	Trauma SICU (TSICU)	01:13	30.09%
Christopher Green	87	Female	Unknown	Coronary Care Unit (CCU)	07:07	43.14%
Nancy Adams	75	Male	White	Surgical Intensive Care Unit (SICU)	04:43	31.81%
Betty Nelson	74	Female	Hispanic/Latino - D...	Medical Intensive Care Unit (MICU)	13:24	62.20%

ACKNOWLEDGEMENTS

- Kyle Michael Shannon
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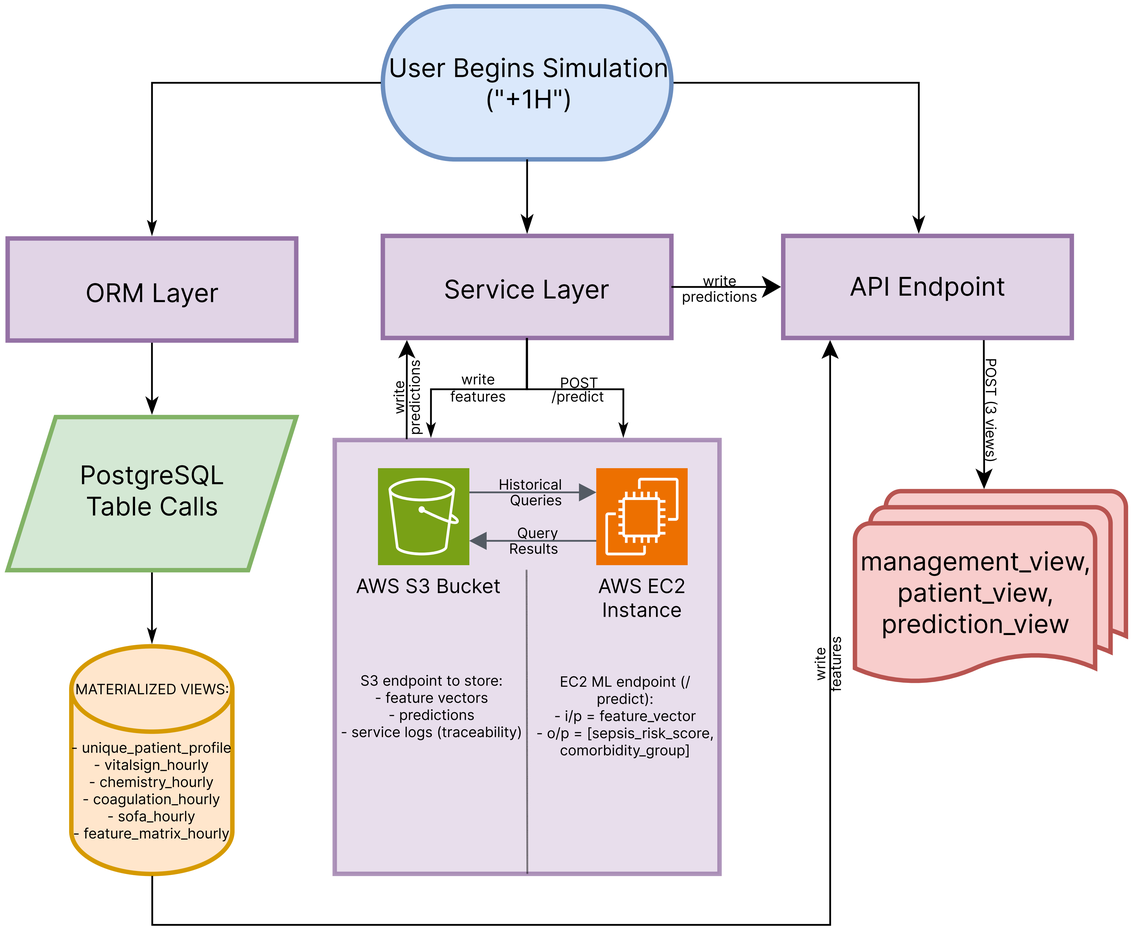
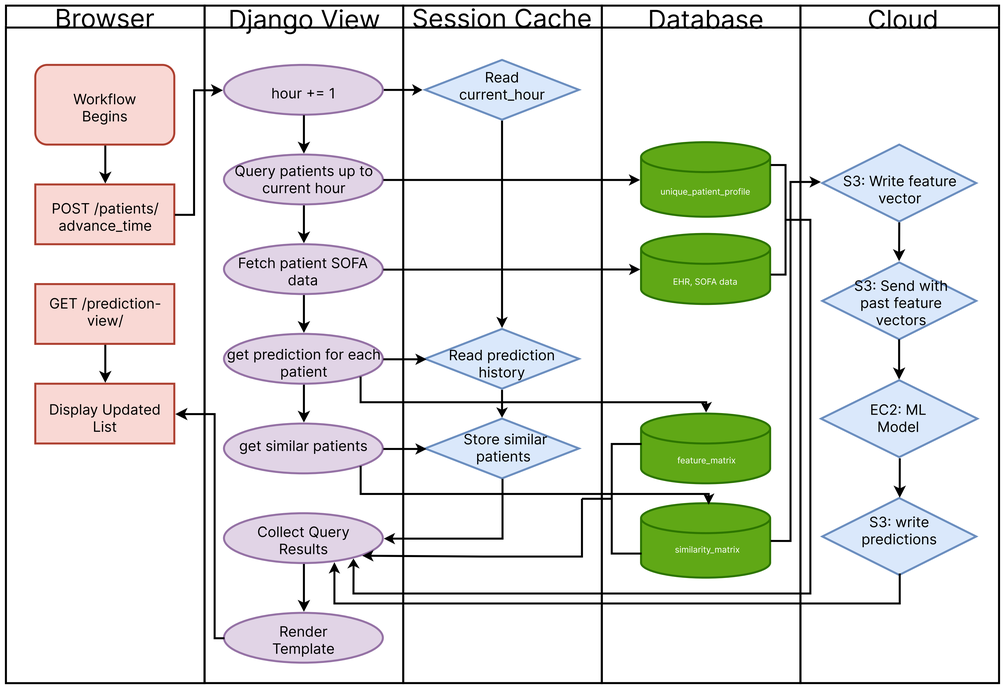
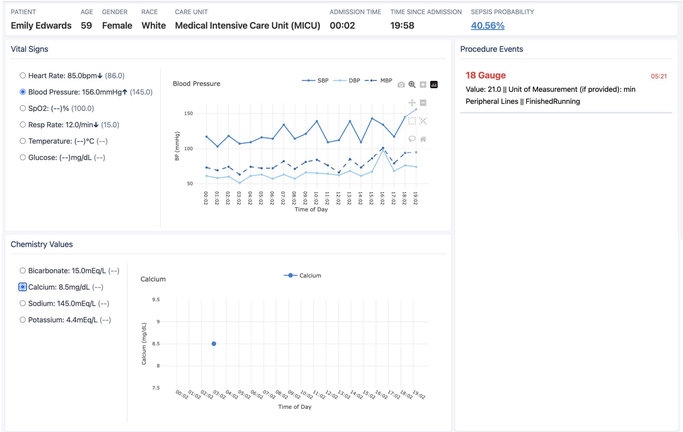
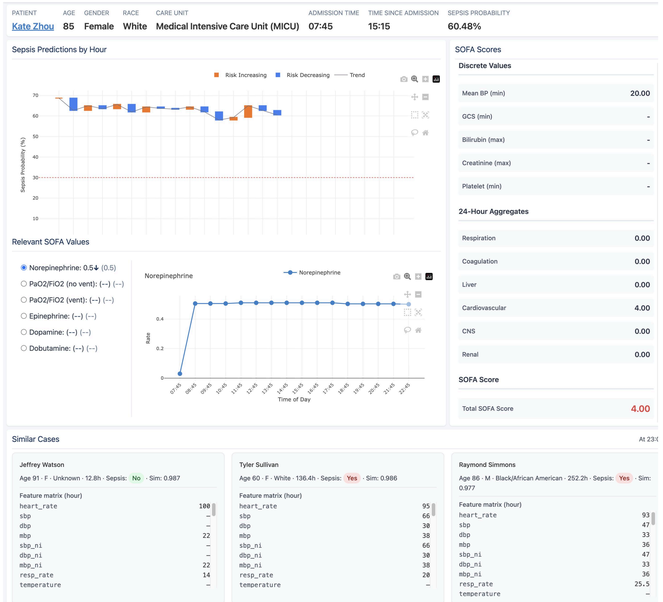




PATIENT VIEW



PREDICTION VIEW



FUTURE STEPS

- Resize the components of each view based on **significance** for each feature
- Change similarity patients calculation based on **weighted** features
- Setup auto-updating processes for production
- Incorporate chest x-rays from **MIMIC-CXR** to improve accuracy of the machine learning model